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University of Plymouth

MODULE CODE: ELEC347 Exam

**TITLE OF PAPER: Information and Communication Signal
Processing**

TIME ALLOWED 3 Hours

DATE Friday 17th January 2025

TIME 09:00 – 12:00

FACULTY Science and Engineering

SCHOOL Engineering, Computing and Mathematics

ACADEMIC YEAR 2024 / 2025

STAGE Final

INSTRUCTIONS TO CANDIDATES:

The paper consists of **TWO** Sections.

Section A: Attempt **ALL TEN** questions in this Section.

Section B: Candidates should attempt any **TWO** questions from the FIVE available in this section. All questions in this Section carry equal marks. Marks for parts of questions are shown where appropriate.

Candidates are not permitted to look at the examination paper until instructed to do so.

SECTION A - Attempt ALL questions in this Section.

1. State **TWO** main methods of controlling the level of aliasing error in real-time DSP systems in practice.

[4 marks]

2. A digital signal processing system, with analogue audio signal input in the range 0-10 kHz, uses over-sampling techniques to convert the analogue signal into digital at a rate of 3.072 MHz. Determine the reduction in the quantisation noise power, in dB, made possible by oversampling.

[6 marks]

3. (a) Why is bit reversal operation on the input sequence necessary in FFT?

[5 marks]

- (b) Explain what would happen to the data samples $x(5)$ and $x(10)$ after bit reversal in a 32-point FFT.

[2 marks]

4. An analogue signal to be spectrum analysed is sampled at 500 Hz to produce 2048 data values. What is the frequency spacing between the spectral samples?

[4 marks]

5. What is zero-padding and why is it sometimes used in relation to FFT-based spectrum analysis?

[5 marks]

6. (a) State **TWO** main advantages of FIR filters over IIR filters in practice.
[4 marks]

(b) An FIR lowpass filter with a positive symmetrical impulse response has 120 coefficients. Assume the filter operates at a sampling rate of 8 kHz, determine the time delay through the filter.
[4 marks]

7. A discrete-time filter has two zeros located at $\pm j$ and two poles at $z=\pm 0.8$

(a) Sketch the pole-zero diagram of the filter and deduce its transfer.
[6 marks]

(b) Sketch the magnitude-frequency response between 0 and the sampling frequency. What type of filter is this?
[4 marks]

8. State the four basic components of sample rate converters and the role of each component.
[6 marks]

9. A 2-stage decimator satisfies the following overall specifications:

Input sampling rate, F_s	3072 kHz
Output sampling rate	96 kHz
Frequency band of interest	0 – 20 kHz

Determine:

(a) The decimation factor for each stage.
[4 marks]

(b) The overall decimation factor.
[2 marks]

10. State **TWO** main adverse effects of coefficient quantisation on the performance of fixed-point digital filters in practice.

[4 marks]

END OF SECTION A

SECTION B - Attempt TWO questions ONLY from this Section.

11. (a) The front-end of a real-time DSP system with a wide band input signal is depicted in the **Fig Q11 (on page 6)**.

(i) Explain briefly, in terms of the convolution theorem, the relationship between each of the following:

- $x''(t)$ and $x'(t)$
- $X''(f)$ and $X'(f)$

where $X'(f)$ and $X''(f)$ are the spectrum of the signals $x'(t)$ and $x''(t)$, respectively.

[4 marks]

(ii) Determine the minimum sampling frequency, F_s (min), to give a signal-to-aliasing error level of at least 10: 1 in the passband. State any assumptions made.

[11 marks]

(b) Explain, with the aid of suitable sketches, the oversampling technique and how it can be used to increase the effective resolution of Nyquist rate analogue-to-digital converters.

[5 marks]

(c) A digital audio system uses the oversampling technique and an 10-bit, Nyquist rate converter to digitise an analogue input signal which has frequency components in the range 0-5kHz. Estimate the effective resolution, in bits, of the converter if the sampling rate is 32 MHz. Show clearly how you obtained your answer. Comment on the practical problems associated with this approach.

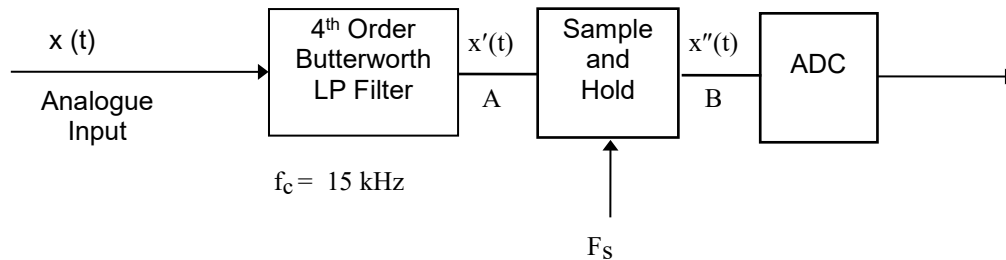
You may assume the following: $\frac{2f_{max}}{F_s} = 2^{-(B_2-B_1)}$, where F_s is the sampling frequency, f_{max} the highest signal frequency of interest, B_1

(Q11. Continued Over/.....)

(Q11. Continued)

and B_2 the ADC resolutions of the Nyquist rate and oversampled converters, respectively.

[10 marks]



NB: The magnitude response of the Butterworth filter is characterised by the following equation: $|H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^{2N}}}$, where N is the order of the filter.

$$|H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^{2N}}}$$

Fig.Q11.

12. (a) What are the main advantages and disadvantages of the Bilinear Z Transform (BZT) method in the design of digital filters in practice?

[4 marks]

- (b) A requirement exists to design a discrete-time, IIR bandstop filter, with a Butterworth characteristic, meeting the following specifications:

stopband	200 – 300 Hz
sampling frequency	1 kHz
filter order, N	2

- (i) Starting with a transfer function for a suitable prototype, analogue lowpass filter, determine the transfer function and hence the coefficients of the discrete-time filter using the BZT method.

[12 marks]

(Q12. Continued Over/.....)

(Q12. Continued)

- (ii) Determine the locations of the poles and zeros of the prototype lowpass filter, the pre-warped bandstop filter, and the discrete-time filter.

[6 marks]

- (iii) Explain, with the aid of sketches of the pole-zero diagrams, how the poles and zeros of the discrete-time filter in b (ii) have arisen and hence trace their origin to the prototype lowpass filter. Comment on the stability of the discrete-time filter.

[8 marks]

N.B: (1) The lowpass-to-bandstop transformation is given by:

$$s = \frac{Ws}{s^2 + \omega_0^2} \quad \text{lowpass-to-bandstop}$$

$$\text{where } \omega_0^2 = \omega_{p1}' \omega_{p2}', \quad W = \omega_{p2}' - \omega_{p1}'.$$

and $\omega_p' = \tan\left(\frac{\omega_p T}{2}\right)$ is the pre-warped bandedge frequency. Other variables have the usual meaning. The BZT equation is given by:

$$s \rightarrow \frac{z-1}{z+1}$$

(2) The roots of the second order quadratic equation, $ax^2+bx+ c$, are:

$$R1 = [-b + \text{SQRT}(4ac - b^2)]/(2 a)$$

$$R2 = [-b - \text{SQRT}(4ac - b^2)]/(2 a)$$

13. (a) A requirement exists for the design of a 2-stage decimator for an audio multi-rate system that meets the following specifications:

input sampling frequency, F_s	300 kHz
decimating factor, M	30
highest frequency of interest in the data, f_p	4 kHz
passband ripple, δ_p	0.05
stopband ripple, δ_s	0.01

A preliminary exploitation of the design problem, with the aid of a software tool, produced the information summarised in **Table Q13, (on page 9)**. From the design specifications and the information in **Table Q13**.

- (i) Design, at a block diagram level, the 2-stage decimator. Your answer should indicate a suitable pair of decimation factors, with a justification for your choice, and the sampling frequencies at the input and output of each stage of decimation.

[10 marks]

- (ii) For each stage of the decimator, determine the following parameters for the decimating filter:

the bandedge frequencies
the normalised transition width
the stop and passband ripples
filter length.

[7 marks]

NB: You may assume that the stop band for stage i filter is bounded by: $(F_i - F_s / 2M) \leq f \leq F_{i-1} / 2, i = 1, 2, \dots, I$, where F_i is the output sample rate at stage i , F_s is the original sample rate and M is the overall decimation factor.

(Q13. Continued Over/.....)

(Q13. Continued)

(b) The 2-stage decimator in (a) is to be implemented using direct form FIR filters.

(i) Sketch the signal flow graph for a direct implementation of the first stage of decimation. Explain how the implementation works.

[5 marks]

(ii) Sketch the signal flow diagram of a computationally efficient implementation. Explain with the aid of a flowchart how the efficient implementation works and why it is efficient.

[8 marks]

#	M_1	M_2	N_1	N_2	MPS	TSR
0	15	2	43	32	1, 175, 398	75
1	10	3	23	47	1, 148, 777	70
2	6	5	12	79	1, 365, 238	90

Table Q13 A 2-stage decimator design information

Notes

#	Design option number
M_1, M_2	Decimation factor pair for stages 1 and 2.
N_1, N_2	Number of filter coefficients for stages 1 and 2.
MPS	Number of multiplications per second
TSR	Total storage requirements.

14. (a) Describe, with the aid of a diagram, the five design stages for digital filters. Why is the design process iterative? [10 marks]
- (b) Identify the main outcomes of each design stage. [5 marks]
- (c) A requirement exists for a digital notch filter that meets the requirements given below.
- (i) Determine the transfer function, $H(z)$, of the filter using the pole-zero placement method. [10 marks]
- (ii) Starting with the transfer function, $H(z)$ above, derive the difference equation for the filter. [5 marks]

Filter specifications:

Notch frequency	60 Hz
3 dB notch bandwidth	$\pm 5\text{Hz}$
Sampling frequency	1.2 kHz

You may assume that the pole radius, r , is given by:

$$r \approx 1 - \frac{\text{bandwidth} \times \pi}{F_s}$$

15. (a) In relation to radix-2 and radix-4 FFT algorithms write brief notes on each of the following, explaining their meaning and role:
- (i) in-place computation
- (ii) bit-reversal and digital reversal [8 marks]

(Q15. Continued Over/.....)

(Q15. Continued)

- (b) The signal flowgraph for an 8-point decimation in time FFT algorithm is depicted in **Figure Q15, (on page 12)**. Use the flowgraph to compute the DFT coefficients $X(0)$, $X(2)$ and $X(6)$ for the following input data sequence:

n	0	1	2	3	4	5	6	7
x(n)	0	2	0	0	1	1	0	2

[12 marks]

- (c) A real-time spectrum analyser uses a 1024-point FFT processor. If a 64-point FFT takes 20 μs to perform, estimate the maximum sampling frequency that the system can support. Assume a complex multiplication takes twice as long as a complex addition and ignore all other overheads.

[10 marks]

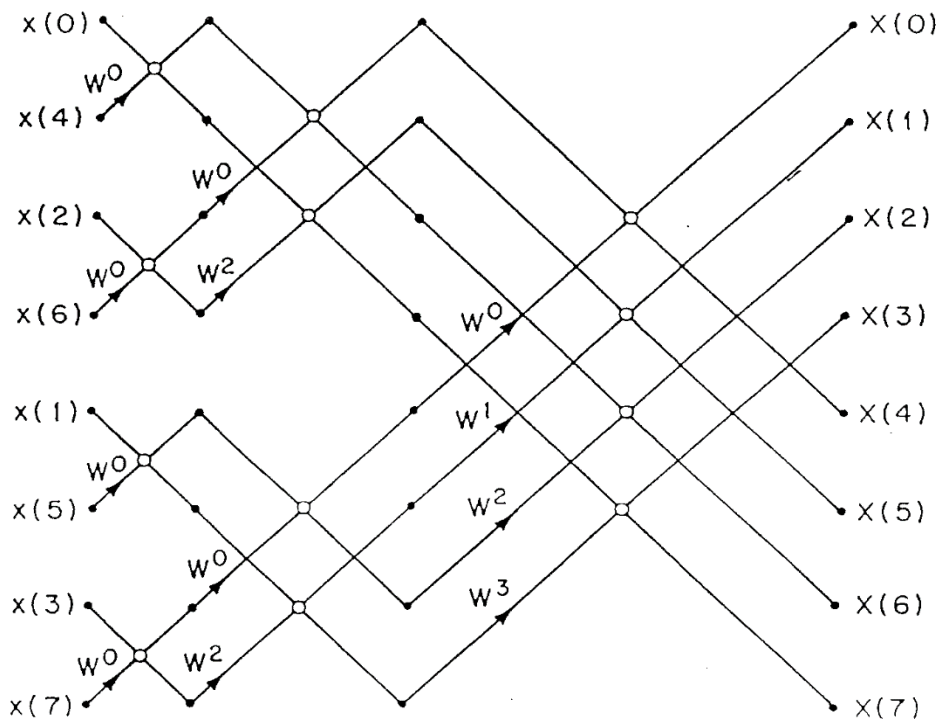
NB: You may assume the following for an N-point radix-2 FFT:

Number of complex multiplications $= \frac{N}{2} \log_2 N$.

Number of complex additions $= N \log_2 N$

(Q15. Continued Over/.....)

(Q15. Continued)



NB:

$$(1) W^0 = 1, W^1 = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j, W^2 = -j, W^3 = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j$$

(2) Each butterfly takes two data samples, A and B, and produces two data points, A' and B': $A' = A + W^k B$; $B' = A - W^k B$

Fig. Q15

END OF PAPER